

Retail Vendor Performance & Profitability Analysis

Introduction

This project analyzes vendor performance, cost efficiency, inventory optimization, and profitability using retail and wholesale data. Python, SQLite, SQL, and Power BI are integrated to transform raw operational data into management-ready insights.

Abstract

The solution builds an end-to-end analytics pipeline that ingests six retail datasets, performs ETL and SQL-based aggregation, engineers business metrics such as Gross Profit, Profit Margin, Stock Turnover, and Sales-to-Purchase Ratio, and presents executive dashboards. Analysis identifies top-performing vendors, underperforming brands, cost inefficiencies, and inventory improvement opportunities.

Tools Used

Python (Pandas, SQLAlchemy, Logging), SQLite, SQL, Power Query, Power BI, DAX, Matplotlib, Seaborn, Jupyter Notebook.

Steps Involved in Building the Project

1. Imported six raw datasets into SQLite using a Python ETL pipeline with logging. 2. Cleaned, transformed, and joined data using SQL to create a consolidated vendor_sales_summary table. 3. Engineered KPIs: Gross Profit, Profit Margin, Stock Turnover, and Sales-to-Purchase Ratio. 4. Performed EDA using Pandas, Matplotlib, and Seaborn to identify trends and validate data. 5. Built two Power BI dashboards: Executive Summary and Cost Efficiency & Inventory. 6. Generated business insights on vendor profitability, freight costs, purchasing efficiency, and inventory optimization.

Conclusion

The project demonstrates an end-to-end analytics workflow from data engineering to executive reporting. It identifies top and underperforming vendors, highlights freight and inventory optimization opportunities, and supports data-driven procurement and profitability decisions. Future enhancements include automated refresh, predictive vendor performance models, and Power BI Service deployment.